STRUCTURED QUERY LANGUAGE(SQL) Practice Questions

By:-
Amit Yerpude
PGT (Computer Science)
Kendriya Vidyalaya , Khagaria

Consider the following tables Product and Client. Write SQL commands for the statement (i) to (iv) and give outputs for SQL queries (v) to (viii)

Table: CLIENT

C_ID	Client Name	City	P_ID
01	TalcomPowder	Delhi	FW05
06	Face Wash	Mumbai	BS01
12	Bath Soap	Delhi	SH06
15	Shampoo	Delhi	FW12
16	Face Wash	Banglore	TP01

Table: PRODUCT

P_ID	Product Name	Manufacturer	Price
TP01	TalcomPowder	LAK	40
FW05	Face Wash	ABC	45
BS01	Bath Soap	ABC	55
SH06	Shampoo	XYZ	120
FW12	Face Wash	XYZ	95

(i) To display the details of those Clients whose city is Delhi.

Ans: Select * from Client where City=”Delhi”

(ii) To display the details of Products who’s Price is in the range of 50 to 100(Both values included).

Ans: Select * from product where Price between 50 and 100

(iii) To display the ClientName, City from table Client, and ProductName and Price from table Product, with their corresponding matching P_ID.

Ans: Select ClientName,City,ProductName, Price from Product,Client where Product.P_ID=Client.P_ID.

(iv) To increase the Price of all Products by 10

Ans: Update Product Set Price=Price +10

(v) SELECT DISTINCT City FROM Client.

Ans: (The above question may consist DISTINCT City. If it is DISTINCT City, the following is the answer)

City
Delhi
Mumbai
Bangalore

(vi) SELECT Manufacturer, MAX(Price), Min(Price), Count(*) FROM Product GROUP BY Manufacturer;

Ans:

Manufacturer	Max(Price)	Min(Price)	Count(*)
LAK	40	40	1
ABC	55	45	2
XYZ	120	95	2

(vii) SELECT ClientName, ManufacturerName FROM Product, Client WHERE Client.P_Id=Product.P_Id;

Ans:

ClientName	ManufacturerName
Cosmetic Shop	ABC
Total Health	ABC
Live Life	XYZ
Pretty Woman	XYZ
Dreams	LAK

(viii) SELECT ProductName, Price * 4 FROM Product.

Ans:

ProductName	Price*4
TalcomPoweder	160
Face Wash	180
Bath Soap	220
Shampoo	480
Face Wash	380

Consider the following tables Item and Customer.
Write SQL commands for the statement (i) to (iv) and give outputs for SQL queries (v) to (viii)

Table: CUSTOMER

C_ID	Cust_Name	City	I_ID
01	N.Roy	Delhi	LC03
06	H.Singh	Mumbai	PC03
12	R.Pandey	Delhi	PC06
15	C.Sharma	Delhi	LC03
16	K.Agarwalh	Banglore	PC01

Table: ITEM

I_ID	ItemName	Manu	Price
PC01	Personal Computer	ABC	35000
LC05	Laptop	ABC	55000
PC03	Personal Computer	XYZ	32000
PC06	Personal Computer	COMP	37000
LC03	Laptop	PQR	57000

(i)To display the details of those Customers whose city is Delhi.

Ans: Select all from Customer Where City="Delhi"

(ii) To display the details of Item whose Price is in the range of 35000 to 55000 (Both values included).

Ans: Select all from Item Where Price>=35000 and Price <=55000

(iii) To display the CustomerName, City from table Customer, and ItemName and Price from table Item, with their corresponding matching I_ID.

Ans: Select CustomerName,City,ItemName, Price from Item,Customer where Item.I_ID=Customer.I_ID.

(iv) To increase the Price of all Items by 1000 in the table Item.

Ans: Update Item set Price=Price+1000

(v) SELECT DISTINCT City FROM Customer.

Ans:

City
Delhi
Mumbai
Bangalore

(vi) SELECT ItemName, MAX(Price), Count(*) FROM Item GROUP BY ItemName;

Ans:

ItemName	**Max(Price)**	**Count(*)**
Personal Computer	37000	3
Laptop	57000	2

(vii) SELECT CustomerName, Manufacturer FROM Item, Customer WHERE Item.Item_Id=Customer.Item_Id;

Ans:

CustomerName	**Manufacturer Name**
N.Roy	PQR
H.Singh	XYZ
R.Pandey	COMP
C.Sharma	PQR
K.Agarwal	ABC

(viii) SELECT ItemName, Price * 100 FROM Item WHERE Manufacturer = 'ABC';

Ans:

ItemName	**Price*100**
Personal Computer	3500000
Laptop	5500000

Consider the following tables Consignor and Consignee. Write SQL command for the statements(i)to(iv)

TABLE : CONSIGNOR

CneeID	CnorID	CneeName	CneeAddress	CneeCity
MU05	ND01	RahulKishore	5,Park Avenue	Mumbai
ND08	ND02	P Dhingr a	16/j,Moore Enclave	New Delhi
KO19	MU15	A P Roy	2A,Central/ avenue	Kolkata
MU32	ND0 2	S mittal	P 245, AB Colony	Mumbai
ND48	MU5 0	B P jain	13,Block d,a,viha	New Delhi

TABLE : CONSIGNEE

CnorID	CnorName	CnorAddress	City
ND01	R singhal	24,ABC Enclave	New Delhi
ND02	AmitKumar	123,Palm Avenue	New Delhi
MU15	R Kohil	5/A,South,Street	Mumbai
MU50	S Kaur	7-K,Westend	Mumbai

(i) To display the names of all consignors from Mumbai.

Ans: Select CnorName from Consignor where city=”Mumbai”;

(ii) To display the cneeID, cnorName, cnorAddress, CneeName, CneeAddress for every Consignee.

Ans: Select CneeId, CnorName, CnorAddress, CneeName, CneeAddress from Consignor,Consignee where Consignor.CnorId=Consignee.CnorId;

(iii) To display the consignee details in ascending order of CneeName.

Ans: Select * from Consignee Orderby CneeName Asc;

(iv) To display number of consignors from each city.

Ans: Select city, count(*) from Consignors group by city;

Study the following tables FLIGHTS and FARES and write SQL commands for the questions (i) to (iv) and give outputs for SQL quires (v) to(vi).

TABLE: FLIGHTS

FL_NO	STARTING	ENDING	NO_ FLGHTS	NO_ STOPS
IC301	MUMBAI	DELHI	8	0
IC799	BANGALORE	DELHI	2	1
MC101	INDORE	MUMBAI	3	0
IC302	DELHI	MUMBAI	8	0
AM812	KANPUR	BANGLORE	3	1
IC899	MUMBAI	KOCHI	1	4
AM501	DELHI	TRIVENDRUM	1	5
MU499	MUMBAI	MADRAS	3	3
IC701	DELHI	AHMEDABAD	4	0

TABLE:FARES

FL_NO	AIRLINES	FARE	TAX%
IC701	INDIAN AIRLINES	6500	10
MU499	SAHARA	9400	5
AM501	JET AIRWAYS	13450	8
IC899	INDIAN AIRLINES	8300	4
IC302	INDIAN AIRLINES	4300	10
IC799	INDIAN AIRLINES	1050	10
MC101	DECCAN AIRLINES	3500	4

(i) Display FL_NO and NO_FLIGHTS from “KANPUR” TO “BANGALORE” from the table FLIGHTS.

Ans: Select FL_NO, NO_FLIGHTS from FLIGHTS where Starting=”KANPUR” AND ENDING=”BANGALORE”

(ii) Arrange the contents of the table FLIGHTS in the ascending order of FL_NO.

Ans: (Try this as an assignment)

(iii) Display the FL_NO and fare to be paid for the flights from DELHI to MUMBAI using the tables FLIGHTS and FARES, where the fare to paid = FARE+FARE+TAX%/100.

Ans: Select FL_NO, FARE+FARE+(TAX%/100) from FLIGHTS, FARES where Starting=”DELHI” AND Ending=”MUMBAI”

(iv) Display the minimum fare “Indian Airlines” is offering from the tables FARES.

Ans: Select min(FARE) from FARES Where AIRLINES=”Indian Airlines”

v) Select FL_NO,NO_FLIGHTS,AIRLINES from FLIGHTS, FARES Where STARTING = “DELHI” AND FLIGHTS.FL_NO = FARES.FL_NO

Ans: FL_NO NO_FLIGHTS AIRLINES IC799 2 Indian Airlines

(vi) SELECT count (distinct ENDING) from FLIGHTS.

Ans: (Try this answer as an assignment)

Study the following tables DOCTOR and SALARY and write SQL commands for the questions (i) to (iv) and give outputs for SQL queries (v) to (vi) :

TABLE: DOCTOR

ID	NAME	DEPT	SEX	EXPERIENCE
101	Johan	ENT	M	12
104	Smith	ORTHOPEDIC	M	5
107	George	CARDIOLOGY	M	10
114	Lara	SKIN	F	3
109	K George	MEDICINE	F	9
105	Johnson	ORTHOPEDIC	M	10
117	Lucy	ENT	F	3
111	Bill	MEDICINE	F	12
130	Murphy	ORTHOPEDIC	M	15

TABLE: SALARY

ID	BASIC	ALLOWANCE	CONSULTAION
101	12000	1000	300
104	23000	2300	500
107	32000	4000	500
114	12000	5200	100
109	42000	1700	200
105	18900	1690	300
130	21700	2600	300

(i) Display NAME of all doctors who are in "MEDICINE" having more than 10 years experience from the Table DOCTOR.

Ans: Select Name from Doctor where Dept="Medicine" and Experience>10

(ii) Display the average salary of all doctors working in "ENT"department using the tables. DOCTORS and SALARY Salary =BASIC+ALLOWANCE.

Ans: Select avg(basic+allowance) from Doctor,Salary where Dept="Ent" and Doctor.Id=Salary.Id

(iii) Display the minimum ALLOWANCE of female doctors.

Ans: Select min(Allowance) from Doctro,Salary where Sex="F" and Doctor.Id=Salary.Id

(iv) Display the highest consultation fee among all male doctors.

Ans: Select max(Consulation) from Doctor,Salary where Sex="M" and Doctor.Id=Salary.Id

(v) SELECT count (*) from DOCTOR where SEX = “F”

Ans: 4

(vi) SELECT NAME, DEPT , BASIC from DOCTOR, SALRY Where DEPT = “ENT” AND DOCTOR.ID = SALARY.ID

Ans:

Name	Dept	Basic
Jonah	Ent	12000

Thank you!!!